

as well as accurate responses like the Live API response. Postman is specially used for testing purposes.

Now, let's look at some good reasons to use Postman.

- *Easily creates test suites:* Here, we can create a collection of integration tests to check whether an API is working as per expectations. In Postman, tests are always run in a specific manner or in a specific order; each and every test is executed after the previous one is finished. In every test, an HTTP request is made and assertions written in JavaScript are then used to verify the integrity of your code.
- Postman stores information for running tests in different environments. Sometimes we write a test and it works perfectly, even when we run it again and again against our local environment. So Postman allows us to store specific data about the different environments and insert the correct environment configuration into our test, automatically.

Jenkins

Jenkins is an automation server. After Jenkins 2.0, Jenkins is no longer just a CI server. Now, there is dedicated focus on CI as well as CD. Some of its features include the following:

- Role based access to users depending on build jobs, folders, pipelines, configurations and actions.
- Integration with Active Directory, LDAP and with the internal database.
- More than 1500 plugins to integrate Jenkins with many open source and commercial tools (refer to <https://plugins.jenkins.io/>).
 - Platform for iOS, .NET, Android and Ruby development.
 - User interface (listview-column plugins).
 - For administration purposes Jenkins has agent controllers, page decorators, users and security, cluster management, CLI extensions, etc.
 - Source code management (SCM connections, SCM related).
 - Build management (build triggers, wrappers, tools, reports, parameters and notifiers; deployment plugins, clean-up actions and artifact uploaders).
- Master agent architecture to manage Jenkins effectively.
- Build pipeline plugins to orchestrate different activities in application life cycle management.
- Pipeline as code/multibranch pipeline to manage pipelines easily in repositories.
 - Blue Ocean – a new user interface to create pipeline as code in a easier way.
 - A new UI for the Jenkins dashboard that is easy to use.
 - Multiple views supported along with folders to manage jobs in an easier manner.
 - Jenkins installation is supported for Docker, FreeBSD, Gentoo, MacOS X, OpenBSD, openSUSE, Red Hat, Fedora, CentOS, Ubuntu/Debian, Windows, and generic Java packages (.war).

Jenkins has more than 155,000 active installations and approximately 1.5 million users. It is a multiple-award-winning open source project.

Slack

Slack has become one of the most popular collaboration tools in recent times. It has a free version, a standard version (US\$ 6.67 per active user, per month billed annually; or US\$ 8 billed monthly), and the Plus version (US\$ 12.50 per active user, per month billed annually; or US\$ 15 billed monthly). It comes with a vast array of compliance certifications and regulations, two of which are:

- FedRAMP (Li-SaaS) Federal Risk and Authorisation Management Program
 - NIST 800-171 - Protecting Controlled Unclassified Information in Non-federal Systems and Organisations
- Its features include:
- Online collaboration
 - Easy integration with automation tools such as Jenkins. Slack notifications can be configured for successful static code analysis, build, deployment, and so on, as well as for when these tasks/processes fail.
 - Workspace is created so that team members communicate and collaborate.
 - Multiple roles, which include workspace owners, workspace admins, members and guests.
 - Slack has channels where team members do messaging, use tools, and transfer files. Channels are used for team communication and collaboration. These include public channels, private channels and shared channels (beta).

There is a Slack app for Android and iOS devices, a Slack dashboard application for computers, and a Web app to use Slack effectively when on the go. **END** 

References

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